



STM32F3 Technical Training

For reference only

Refer to the latest documents for details





Inter-Integrated Circuit (I2C)

I2C Features (1/2)

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- I2C specification rev03 compatibility
- SMBus 2.0 HW support
- PMBus 1.1 Compatibility
- Multi Master and slave capability
- Controls all I²C bus specific sequencing, protocol, arbitration and timing
- Standard, fast and **fast mode + I²C mode (up to 1MHz)**

I2C Features (2/2)

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- 7-bit and 10-bit addressing modes
- **Multiple** 7-bit Addressing Capability with **configurable mask**
- **Programmable setup and hold time**
- **Easy to use event management**
- **Programmable analog and digital noise filter**
- **Wakeup from STOP mode on address match**
- Optional clock stretching
- **Independent clock**

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I2C SDA and SCL noise filter

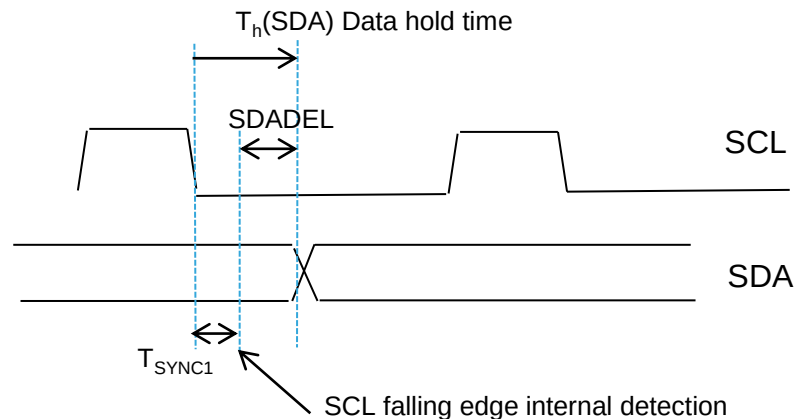
- Analog noise filter in SDA and SCL I/O
 - Can filter spikes with a length up to 50ns
 - This filter can be enabled or disabled by SW (enabled by default)
- Digital noise filter for SDA and SCL
 - Suppress spikes with a programmable length from **0 to 15 I2CCLK periods**.
- Only analog filter can be enabled when Wakeup from STOP feature is enable.
- Filters configuration must be programmed when the I2C is disable.

I2C Programmable timings

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- Setup and Hold timings between SDA and SCL in transmission are programmable by SW with PRESC, SDADEL and SCLDEL fields in I2C Timing Register (I2Cx_TIMINGR).
 - SDADEL is used to generate Data Hold time. $T_{SDADEL} = SDADEL * (PRESC+1) * T_{I2CCLK}$
 - SCLDEL is used to generate Data Setup time. $T_{SCLDEL} = (SCLDEL+1) * (PRESC+1) * T_{I2CCLK}$
- Example Data Hold Time :**



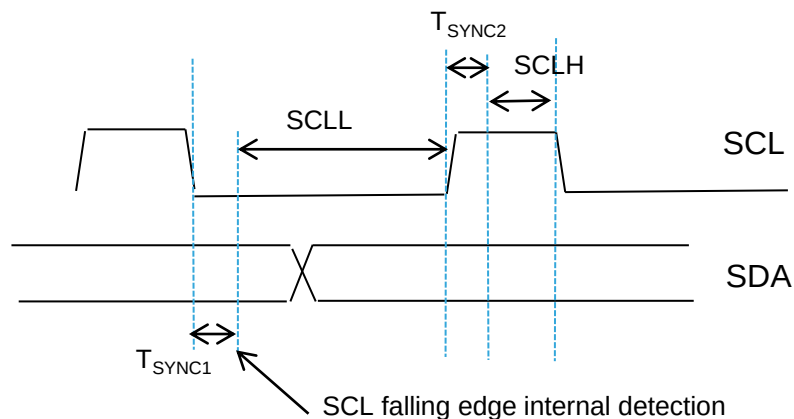
- The Setup and Hold configuration must be programmed when the I2C is disable.
- I2C_Timing_Config_Tool will be available to calculate I2C_TIMINGR value for your application.

I2C Master clock generation

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- SCL Low and High duration are programmable by SW with PRESC, SCLL and SCLH fields in I2C Timing Register (I2Cx_TIMINGR).
 - SCL Low counter is $(SCLL+1) * (PRESC+1) * T_{I2CCLK}$. It starts counting after SCL falling edge internal detection. After counting, SCL is released.
 - SCL High counter is $(SCLH+1) * (PRESC+1) * T_{I2CCLK}$. It starts counting after SCL rising edge internal detection. After counting SCL is driven low.
- The total SCL period is :
 - $T_{SYNC1} + T_{SYNC2} + [(SCLL+1) + (SCLH+1)] * (PRESC+1) * T_{I2CCLK}$
- **SCL Period:**



- The SCLL and SCLH configuration must be programmed when the I2C is disable.
- I2C_Timing_Config_Tool is available in FWLib to calculate I2C_TIMINGR value for your application.

Slave Addressing Mode

- I2C can acknowledge several slave addresses. 2 address registers :
 - I2Cx_OAR1 : 7-bit or 10-bit mode.
 - I2Cx_OAR2 : 7-bit mode only. OA2MSK[2:0] allow to mask from 0 to 7 LSB of OAR2 :

OA2MSK[2:0]	Address match condition
000	address[7:1] = OA2[7:1]
001	address[7:2] = OA2[7:2] (Bit 1 is don't care)
010	address[7:3] = OA2[7:3] (Bit 2:1 are don't care)
...	
111	All addresses are acknowledged except I2C reserved addresses.

Wakeup from STOP on address match

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- When I2CCLK clock is HSI, the I2C is able to wakeup MCU from STOP when it receives its slave address. All addressing mode are supported.
 - During STOP mode and no address reception : HSI is switched off.
 - On START detection, I2C enables HSI, used for address reception.
- Wakeup from STOP is enabled by setting WUPEN in I2C1_CR1.
- Clock stretching must be enabled to ensure proper operation: NOSTRETCH=0.

Easy Master mode management

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- For payload ≤ 255 bytes : **only 1 write action needed !!** (apart data rd/wr)

I2Cx_CR2 is written w/ :

- START=1
- SADD : slave address
- RD_WRN : transfer direction
- NBYTES = N : number of bytes to be transferred
- **AUTOEND** =1 : STOP automatically sent after N data.

AUTOEND

0 : Software end mode	End of transfer SW control after NBYTES data transfer : • TC flag is set. Interrupt if TCIE=1. • TC is cleared when START or STOP is set by SW ➤ If START=1 : RESTART condition is sent
1 : Automatic end mode	STOP condition sent after NBYTES data transfer

- Data transfer managed by Interrupts (TXIS / RXNE) or DMA

Easy to use event management 12

- For payload > 255 : in addition, **RELOAD** must be set in I2Cx_CR2.

RELOAD	
0 : No reload	NBYTES data transfer is followed by STOP or ReSTART
1 : Reload mode	NBYTES is reloaded after NBYTES data transfer (data transfer will continue) : • TCR flag is set. Interrupt if TCIE=1. • TCR is cleared when I2Cx_CR2 is written w/ NBYTES≠0

- AUTOEND = 0 has no effect when RELOAD is set

- By default : I2C slave uses clock stretching .
 - This can be disabled by setting **NOSTRETCH=1**
- Reception : Acknowledge control can be done on selected bytes in **Slave Byte Control (SBC)** mode with **RELOAD=1**
 - **SBC = 1** enables the **NBYTES** counter in slave mode (Tx and Rx modes).
 - **SBC = 1** is allowed only when **NOSTRETCH=0**.

SBC	
0 : Slave Byte Control disable	All received bytes are acknowledged.
1 : Slave Byte Control enable	If RELOAD=1 , after NBYTES data are transferred : • TCR set & SCL stretched before ACK pulse in reception. • TCR is cleared when I2Cx_CR2 is written w/ NBYTES≠0 ➤ if I2Cx_CR2/NACK = 1: received byte is NOT Acknowledged

Interrupt event	Interrupt flag
Receive Buffer Not Empty	RXNE
Transmit buffer Interrupt Status	TXIS
Stop detection interrupt flag	STOPF
Transfer Complete Reload	TCR
Transfer Complete	TC
Address matched	ADDR
NACK reception	NACKF

- **ARP** (Address resolution protocol) support : Device default address, Arbitration in slave mode
- **Host Notify** protocol support : host address
- **Alert** support : Alert pin and Alert Response support
- Configurable **Timeout** detection : Clock low timeout, Cumulative clock low extend time
- Configurable **bus idle** detection
- Command and data **acknowledge control** in SBC mode

SMBUS : PEC (Packet Error Checking)

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- HW PEC calculation is enabled when **PECEN**=1 in I2C1_CR1.
- NBYTES (data transfer counter) is used to :
 - automatically check PEC in reception, after NBYTES-1 are received.
 - Automatic NACK sending in case of failure
 - automatically send PEC in transmission, after NBYTES-1 are sent.
 - Therefore SBC must be set in Slave mode to enable NBYTES counter
- Automatic PEC sending/checking is done when **PECBYTE**=1 in I2C1_CR2.

Error conditions

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Interrupt event	Interrupt flag
Bus error detection	BERR
Arbitration Loss	ARLO
Over-run / Under-run error	OVR
SMBUS : PEC error	PECERR
SMBUS : Timeout error	TIMEOUT
SMBUS : Alert pin detection	ALERT

- How many I2C are in the STM32F3x microcontroller ?

- What are the different I2C modes supported by the I2C1 and I2C2?

- What is the SW sequence to read 3 data from slave address 0xAA in master mode?

- What are the error flags of the I2C?



Thank you !

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